

Lower warning limit

$$LWL = T - k_2 \frac{\hat{\sigma}}{\sqrt{J}} \quad (7.13)$$

Lower control limit

$$LCL = T - k_1 \frac{\hat{\sigma}}{\sqrt{J}} \quad (7.14)$$

The limits depend on a dispersion parameter, traditionally denoted $\hat{\sigma}$ and two coverage factors denoted k_1 and k_2 . In Section 4.4, by convention, fixed factors were applied $k_1 = 3$ and $k_2 = 2$. Different methods are used to define $\hat{\sigma}$ and consequently, various control charts can be obtained:

Method	$\hat{\sigma}$ values
Standard deviation of the preliminary study	s
Standard deviation of repeatability of a preliminary study	s_r
The known standard uncertainty of T (applicable to CRM)	$u(T)$
Predefined percentage c of the target value	$T \times c$

The ALBUMIN example is based on the standard deviation of repeatability. The formula for calculating this standard deviation is presented in Section 3.2 and should be referred to for notations. In addition, Resource E is an Excel worksheet for calculating it. It is summarized as follows:

Standard deviation of repeatability

$$s_r = \sqrt{\frac{\sum_{i=1}^I \sum_{j=1}^J (Z_{ij} - \bar{Z}_i)^2}{I(J-1)}} = \sqrt{\frac{\sum_{i=1}^I SCE_i}{I(J-1)}} \quad (7.15)$$

Finally, the coverage factors can take on various values, but it is typically recommended to take:

Type of limit	Coverage factor	Probability of coverage
Control	$k_1 = 3$	99%
Warning	$k_2 = 2$	95%

7.3.2 Statistical Dispersion Intervals and Control Charts

Using the same dataset, it is possible to calculate the β -ETI, and β - γ -CTI described in Section 5.3 to quickly obtain a Shewhart control chart of the mean. The data may have been collected as part of a validation study or independently in a preliminary study with replicates. This allows us to get a MU estimate from QC measurement values.

Everything below is only applicable if the data collection method for the preliminary study is the same as it will be afterwards. This means that if the QC

Table 7.4 ALBUMIN – simulated results of the preliminary study (total polymer contents expressed in g/100 ml).

Series	Replicate 1	Replicate 2
Series 1	4.18	3.51
Series 2	4.60	5.26
Series 3	4.68	4.40
Series 4	5.20	4.60
Series 5	3.99	5.25
Series 6	4.12	4.68
Series 7	4.75	4.25
Series 8	5.02	3.99
Series 9	3.85	5.00
Series 10	4.11	5.22

size is $J = 2$, the same number of replicates must be routinely collected. The control procedure in place amounts to analyzing size QC $J = 2$ at each date for the control of the method.

In the original document, the estimation of the target value is not presented. For pedagogical reasons, the results of a preliminary study were simulated using a normal law simulator available with Python. The study was supposed to last 10 days, $I = 10$ sets of $J = 2$ measurements whose values are collected in Table 7.4. The estimate of the target value was taken as $T = \bar{\bar{Z}}$ and the standard deviation of repeatability $\hat{\sigma} = s_r$.

As discussed in Section 5.4.4, the β - γ -content tolerance interval (β - γ -CTI) is considered as first choice to set the warning (LWL and UWL) and control (LCL and UCL) limits of the future control chart. The next step is to define the coverage probability β and the confidence level γ . For this, the Food and Drug Administration (FDA) recommendations for biological assays could be considered [24]:

- The trueness should be in the $\pm 20\%$ range, which allows $\beta\% = 80\%$.
- Then, $2/3$ (67%) of the QCs (i.e. 4/6) must fall between the warning limits, which gives $\gamma\% = 67\%$.
- Finally, as it is traditional, the control limits not to be exceeded must correspond to a confidence level $\gamma\% = 99\%$.

Finally, two β - γ -CTIs are computed having probabilities:

	Content probability β	Confidence level γ
Warning limits	0.80	0.67
Control limits	0.80	0.99

Table 7.5 ALBUMIN – preliminary study.

Parameter		Warning	Control
Coverage probability	β	0.80	0.80
Confidence level	γ	0.67	0.99
Grand mean	$\bar{\bar{Z}} = T$	4.533	
Std. dev. of repeatability	s_r	0.5954	
Between-series standard deviation	s_B	0.0000	
Standard deviation of β - γ -CTI	s_{IC}	0.6476	
Coverage factor	k_{IC}	1.303	
Lower limit	L	3.69	3.26
Upper limit	U	5.38	5.80

Parameters of the β - γ -CTI to define limits of the control chart (g/100 ml).

The data from Table 7.4 were copied into Resource I, which provides the limits reported in Table 7.5. It is interesting to check whether the calculation of these limits, carried out classically, i.e. described in the standards and summarized by the Eqs. (7.11) and (7.14), give comparable results. For this purpose, the coverage factors $k_1 = 3$ and $k_2 = 2$, the mean as target value T and, standard deviation of repeatability provided by Table 7.5 as an estimator $\hat{\sigma}$.

The standardized computation gives:

Formula	Result
$LCL = T - k_1 \frac{\hat{\sigma}}{\sqrt{J}}$	$LCL = 4.533 - 3 \times \frac{0.5954}{\sqrt{2}} = 3.27$
$LWL = T - k_2 \frac{\hat{\sigma}}{\sqrt{J}}$	$LWL = 4.533 - 2 \times \frac{0.5954}{\sqrt{2}} = 3.69$
$UWL = T + k_2 \frac{\hat{\sigma}}{\sqrt{J}}$	$UWL = 4.533 + 2 \times \frac{0.5954}{\sqrt{2}} = 5.37$
$UCL = T + k_1 \frac{\hat{\sigma}}{\sqrt{J}}$	$UCL = 4.533 + 3 \times \frac{0.5954}{\sqrt{2}} = 5.80$

With these new limits and the old ones rounded to two decimals, it is easy to check that similar values are obtained. They are plotted in Figure 7.4. This result is not surprising since the FDA recommendations are based on the same statistical principles as those that led to the development of the control charts. For the QC of commercial albumin solutions, the reference solution was used for three and a half years on 34 batches of bovine albumin. Figure 7.4 shows the Shewhart control chart obtained from these data with the limits previously calculated, while QC measurement values, with their dates, are recorded in Table 7.6.

The purpose here is not to comment on the QC problems of the analytical method exhibited by this chart but to explain how to use these 68 measurements to infer the MU of the reference solution and, subsequently, any sample close to that level.

Table 7.6 ALBUMIN – QCs over two years expressed in g/100 mL

Date	Rep.1	Rep.2	Date	Rep.1	Rep.2	Date	Rep.1	Rep.2
10 February 2015	4.70	4.77	27 August 2015	4.11	4.21	19 October 2016	5.17	5.00
23 March 2015	4.62	4.80	14 September 2015	4.61	4.59	9 November 2016	4.98	5.12
13 April 2015	4.83	4.91	4 November 2015	4.95	4.91	12 December 2016	5.29	5.45
23 April 2015	5.43	5.58	24 November 2015	4.94	5.03	11 January 2017	5.49	5.46
19 May 2015	5.30	5.47	17 December 2015	4.89	4.99	17 January 2017	5.31	5.27
28 May 2015	5.51	5.53	2 February 2016	4.67	4.79	24 January 2017	5.31	5.33
1 June 2015	4.93	5.05	24 February 2016	4.58	4.41	2 February 2017	5.33	5.28
29 June 2015	5.06	5.02	23 May 2016	5.11	5.22	14 March 2017	4.88	4.91
8 July 2015	4.70	4.83	8 June 2016	5.11	5.26	3 May 2017	4.01	3.89
3 August 2015	4.17	4.24	8 August 2016	5.20	5.33	16 May 2017	3.63	3.50
25 August 2015	4.14	4.20	7 September 2016	5.17	5.34	21 June 2017	5.30	4.89
						1 August 2017	4.92	4.85

Source: Adapted from EDQM-OMCL [11], Table 2.1.

7.3.3 Estimation of the Reference Value Uncertainty

To obtain the MU of the mean value of the reference albumin solution, the proposal is to calculate the β -ETI with a coverage probability $\beta\% = 95\%$ so that it conforms with the GUM coverage factor. The Resource H Excel worksheet can do this computation after adapting it to the increased number of series. The question is to apply this method to 10 duplicates collected during the preliminary study or to the 34 series of routine QC.

Computed parameters for both datasets are summarised in Table 7.7. Some measures show that the method can be out of control and, therefore, significantly influence the MU estimate. Still, it is not a matter of eliminating them because they also contribute in the MU of the reference solution.

Table 7.7 ALBUMIN – different estimation approaches for the MU of the reference albumin solution (g/100 ml).

Parameter	Symbol	Preliminary study	Quality controls	Approach B
Number of data	$I \times J$	20	68	68
Coverage probability		95%	95%	95%
Grand mean	T	4.53	4.91	4.91
Std. dev. of repeatability	s_r	0.595	0.092	
Between series std. dev.	s_B	0	0.462	
Standard uncertainty	$u_c(T)$	0.610	0.478	0.467
Number of df	N_E	18.9	34.3	
Coverage factor	k_{IT}	2.09	2.03	2
Coverage interval		[3.26, 5.81]	[3.94, 5.88]	[3.97, 5.84]
Relative uncertainty	$UR\%(T)$	28.2%	19.8%	19.0%

It is interesting to compare these results with those obtained by applying approach B and the law of propagation of uncertainty, as described in several guides reported in Table 7.1. To account for the fact that the number of replicates is 2 for each control and using the notations already employed, the combined standard uncertainty of the measurand is then:

$$u_c(T) = \sqrt{\frac{s_r^2}{2} + s_B^2}$$

$$u_c(T) = \sqrt{\frac{0.092^2}{2} + 0.462^2} = 0.467 \text{ g/100 ml}$$

The values obtained are presented in Table 7.7 and lead to close estimates, except for the preliminary study. The total number of measures has a strong influence on the MU estimation. It can be concluded that the relative uncertainty of the polymer content of an albumin solution with a concentration of about 4.9 g/100 ml is between $\pm 19\%$ and $\pm 20\%$.

The use of β -ETI and β - γ -CTI presents the clear advantage of being simple and straightforward, on the one hand, to establish the parameters of a control chart, on the other hand, to estimate the measurement uncertainty of the reference material when the number of collected data is sufficient. After a certain number of QCs, it is also necessary to recalculate and modify the target value of the reference material and the limits of the control chart.

It is also possible to construct control charts at different concentration levels, allowing the relationship between MU and concentration to be estimated. One possible strategy for selecting the reference material is to exploit materials used during the validation procedure. They can be turned into reference materials for several control charts.